

Problem 1-1

Given matrices $A \in \mathbb{C}^{n \times n}$, $B \in \mathbb{C}^{n \times m}$, and a Hermitian matrix $\Lambda \in H^{(n+m)}$. Prove the equivalence of the following two conditions:

1. For all $\omega \in \mathbb{R}$,

$$\begin{bmatrix} (i\omega I - A)^{-1}B \\ I \end{bmatrix}^* \Lambda \begin{bmatrix} (i\omega I - A)^{-1}B \\ I \end{bmatrix} < 0. \quad (1)$$

2. There exists a Hermitian matrix $P \in H^n$ such that

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^* \begin{bmatrix} 0 & P \\ P & 0 \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} + \Lambda < 0. \quad (2)$$

(1 Mark)

Problem 1-2

Given matrices $A \in \mathbb{C}^{n \times n}$, $B \in \mathbb{C}^{n \times m}$, and a Hermitian matrix $\Lambda \in H^{(n+m)}$. Prove the equivalence of the following two conditions:

1. For all $\omega \in [\omega_1, \omega_2]$,

$$\begin{bmatrix} (i\omega I - A)^{-1}B \\ I \end{bmatrix}^* \Lambda \begin{bmatrix} (i\omega I - A)^{-1}B \\ I \end{bmatrix} < 0. \quad (3)$$

2. There exist Hermitian matrices $P, Q \in H^n$, with $Q \geq 0$, such that

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^* \begin{bmatrix} -Q & P + i\omega_c Q \\ P - i\omega_c Q & -\omega_1 \omega_2 Q \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} + \Lambda < 0, \quad (4)$$

where

$$\omega_c = \frac{\omega_1 + \omega_2}{2}. \quad (5)$$

(2 Marks)

Problem 1-3

Given matrices $A \in \mathbb{C}^{n \times n}$, $B \in \mathbb{C}^{n \times m}$, and a Hermitian matrix $\Lambda \in H^{(n+m)}$, where A is a Metzler matrix and B is a nonnegative matrix. Prove the equivalence of the following two conditions:

1. For all $\omega \in \mathbb{R}$,

$$\begin{bmatrix} (i\omega I - A)^{-1}B \\ I \end{bmatrix}^* \Lambda \begin{bmatrix} (i\omega I - A)^{-1}B \\ I \end{bmatrix} < 0. \quad (6)$$

2. There exists a diagonal Hermitian matrix $P \in H^n$, with $P > 0$, such that

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^* \begin{bmatrix} 0 & P \\ P & 0 \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} + \Lambda < 0. \quad (7)$$

(2 Marks)

Problem 1-4

For Bonus Problem 1-2, when A is a Metzler matrix and B is a nonnegative matrix, please figure out whether there is any special structure of the matrices P, Q —without introducing conservativeness—to the inequality

$$\begin{bmatrix} A & B \\ I & 0 \end{bmatrix}^* \begin{bmatrix} -Q & P + i\omega_c Q \\ P - i\omega_c Q & -\omega_1\omega_2 Q \end{bmatrix} \begin{bmatrix} A & B \\ I & 0 \end{bmatrix} + \Lambda < 0, \quad (8)$$

in condition (2) of Bonus Problem 1-2, where

$$\omega_c = \frac{\omega_1 + \omega_2}{2}. \quad (9)$$

(5 Marks)